

NAGAPATTINAM, India

WMO: 433470

Lat: 10.767N

Lon: 79.850E

Elev: 30

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 71.7	(c) 73.1	(d) 63.3	(e) 87.1	(f) 80.7	(g) 65.1	(h) 93.0	(i) 80.6	(j) 12.3	(k) 79.8	(l) 11.7	(m) 80.0	(n) 4.2	(o) 320	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	15.3	99.9	79.2	98.4	79.1	96.8	79.2	85.0	92.2	84.3	91.2	83.6	90.3	7.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.4	174.5	88.8	82.6	170.1	88.5	81.9	166.1	88.0	49.5	93.1	48.6	91.4	47.8	90.4	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.5	(b) 11.9	(c) 10.6	(e) 69.1	(f) 105.0	(g) 1.9	(h) 3.2	(i) 67.8	(j) 107.3	(k) 66.7	(l) 109.2	(m) 65.6	(n) 111.0	(o) 64.3	(p) 113.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.9	79.1	80.8	84.4	88.2	89.4	89.9	88.9	87.1	86.5	84.0	80.8	79.3	
	DBStd	4.55	1.97	2.08	2.06	2.00	2.98	2.70	2.20	2.88	2.49	2.99	2.36	2.22	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	12730	902	862	1067	1145	1221	1196	1205	1149	1094	1055	925	909	
	CDD65	7255	437	442	602	695	756	746	740	684	644	590	475	444	
	CDH74	88114	3676	4182	7150	9716	10609	10384	10041	8785	8120	6913	4675	3863	
	CDH80	40794	559	1031	2911	5405	6189	6109	5618	4494	3944	2781	1122	631	
Wind	WSAvg	4.6	4.4	3.8	3.4	4.1	5.6	6.5	5.8	5.2	4.6	3.2	4.0	4.7	
Precipitation	PrecAvg	48.9	1.5	0.7	0.8	1.1	1.9	1.5	1.6	3.1	3.3	8.6	16.1	9.0	
	PrecMax	69.7	7.8	3.6	7.3	8.9	7.4	6.1	5.0	9.5	8.0	29.0	33.2	18.0	
	PrecMin	24.6	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.4	0.0	0.0	0.7	1.6	
	PrecStd	11.4	1.7	1.1	1.7	1.9	1.9	1.8	1.2	2.1	1.9	6.0	8.1	6.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.3	89.5	94.0	99.4	102.1	102.0	100.5	99.7	98.4	95.1	89.9	86.6	
		MCWB	76.9	79.0	79.8	80.9	79.2	80.1	78.9	78.0	77.0	78.9	79.2	77.9	
	2%	DB	85.0	87.7	91.6	95.7	99.3	99.8	98.4	97.4	95.6	92.5	87.7	85.3	
		MCWB	76.2	78.2	80.3	82.1	80.3	79.1	78.4	77.9	77.9	79.7	79.2	77.7	
	5%	DB	84.0	86.3	90.4	94.0	97.1	98.1	96.7	95.3	93.5	90.7	86.2	84.3	
		MCWB	75.5	77.0	79.9	82.3	81.2	79.2	78.5	78.5	78.8	80.0	79.0	77.3	
	10%	DB	82.9	85.1	89.1	92.6	94.9	96.3	94.8	93.2	91.8	89.1	84.8	83.2	
		MCWB	74.9	76.4	79.4	82.0	82.3	79.2	78.9	79.0	79.7	80.0	78.6	76.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.9	80.8	83.4	85.3	86.7	85.7	84.6	84.9	83.6	83.3	81.9	81.0
			MCDB	84.4	86.9	90.2	93.7	93.1	94.6	91.5	91.1	90.1	89.6	86.2	83.9
2%		WB	78.5	79.6	81.9	84.4	85.2	84.2	83.3	83.4	82.6	82.1	81.0	79.9	
		MCDB	82.6	85.4	88.4	92.3	91.8	91.1	89.8	89.8	89.1	88.0	84.9	83.1	
5%		WB	77.6	78.7	81.1	83.6	84.4	83.1	82.4	82.2	81.8	81.5	80.3	79.1	
		MCDB	81.7	84.4	87.6	91.1	90.9	89.7	88.6	88.6	88.0	87.1	84.0	82.3	
10%		WB	76.7	77.9	80.4	82.8	83.7	82.1	81.6	81.3	81.1	80.9	79.7	78.3	
		MCDB	80.9	83.4	86.8	90.0	90.2	88.6	88.1	87.5	87.1	86.2	83.3	81.4	
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.0	10.5	10.6	12.9	15.3	14.5	14.1	13.0	10.0	7.5	7.2	
		MCDBR	9.6	10.3	11.6	11.9	15.5	17.0	16.4	16.8	15.1	12.5	9.5	8.5	
		MCWBR	4.4	4.3	4.1	4.0	5.1	5.6	5.5	5.2	4.8	4.3	3.9	4.2	
	5% WB	MCDBR	8.2	9.2	10.4	11.1	12.6	15.6	14.8	13.6	12.7	10.8	8.1	7.3	
		MCWBR	4.6	4.3	4.4	4.6	5.1	6.3	6.1	6.1	5.3	4.6	4.3	4.5	
Clear-Sky Solar Irradiance	taub	0.455	0.446	0.481	0.496	0.581	0.517	0.525	0.514	0.502	0.519	0.503	0.494		
	taud	2.257	2.271	2.178	2.193	1.955	2.161	2.131	2.169	2.205	2.151	2.172	2.161		
	Ebn at Noon	270	277	268	262	236	248	247	253	259	253	254	255		
	Edh at Noon	43	44	49	48	60	48	49	49	47	49	47	47		
All-Sky Solar Radiation	RadAvg	1593	1876	2030	2006	1905	1825	1780	1813	1806	1534	1305	1335		
	RadStd	87	82	94	95	117	104	92	89	107	100	136	111		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.50	N/A	N/A	+1.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page